Intra uterine transfer (IUT)

Embryo with more than 8 blastomeres is inserted into uterus to complete its further development.

Gamete intra-fallopian transfer (GIFT)

Transfer of an ovum collected from a donor into the fallopian tube. In this the eggs are collected from the ovaries and placed with the sperms in one of the fallopian tubes. The zygote travels toward the uterus and gets implanted in the inner lining of the uterus.

Intra-cytoplasmic sperm injection (ICSI)

In this method only one sperm is injected into the focal point of the egg to fertilize. The sperm is carefully injected into the cytoplasm of the egg. Fertilization occurs in 75 - 85% of eggs injected with the sperms. The zygote is allowed to divide to form an 8 celled blastomere and then transferred to the uterus to develop a protective pregnancy.

Surrogacy

Surrogacy is a method of assisted reproduction or agreement whereby a woman agrees to carry a pregnancy for another person, who will become the newborn child's parent after birth. Through *in vitro* fertilization (IVF), embryos are created in a lab and are transferred into the surrogate mother's uterus.

Male infertility

Azoospermia is defined as the absence of spermatozoa in the ejaculate semen on atleast two occasions and is observed approximately in 1% of the population.

Micro-testicular sperm extraction (TESE)
Microsurgical sperm retrieval from the testicle involves a small midline incision in the scrotum, through which one or both testicles can be seen. Under the microscope, the seminiferous tubules are dilated and small amount of testicular tissue in areas of active sperm production are removed and improved for sperm yield compared to traditional biopsy techniques.

3.9. Detection of foetal disorders during early pregnancy

Ultrasound scanning

Ultrasound has no known risks other than mild discomfort due to pressure from the transducer on the abdomen or vagina. No radiation is used during this procedure. Ultrasonography is usually performed in the first trimester for dating, determination of the number of foetuses, and for assessment of early pregnancy complications.

There are several types of ultrasound imaging techniques. As the most common type, the 2-D ultrasound provides a flat picture of one aspect of the baby. The 3-D image allows the health care provider to see the width, height and depth of the images, which can be helpful during the diagnosis. The latest technology is 4-D ultrasound, which allows the health care provider to visualize the unborn baby moving in real time with a three-dimensional image.

Amniocentesis

Amniocentesis involves taking a small sample of the amniotic fluid that surrounds the foetus to diagnose for chromosomal abnormalities (**Fig. 3.1**).

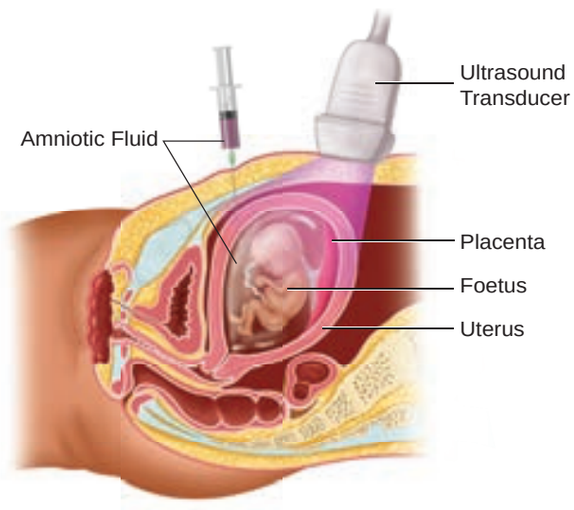


Fig. 3.1 Amniocentesis

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